



## A Web-Based Restaurant Management System with Kitchen Workflow and Inventory Monitoring

**Field:** E-Commerce / Industry

**SDG Alignment:** SDG 8 (Decent Work and Economic Growth) and SDG 9 (Industry, Innovation, and Infrastructure)

### INTRODUCTION

This study focuses on the development of GrabEat: A Web-Based Restaurant Management System with Kitchen Workflow and Inventory Monitoring, designed to improve the efficiency of restaurant operations through a centralized digital platform. The system integrates order processing, kitchen workflow management, and inventory tracking to streamline daily activities within a restaurant environment.

The importance of this study lies in its ability to address common operational challenges faced by restaurants, such as delays in order processing, miscommunication between staff, and poor inventory management. By implementing a web-based system, restaurant businesses can improve productivity, enhance service quality, and reduce errors. This project is relevant to the field of e-commerce and industry, as it applies digital solutions to optimize business processes and supports technological advancement aligned with modern industry needs.

In many restaurants, traditional or manual systems are still used, which often result in inefficiencies such as lost orders, delayed food preparation, and inaccurate stock monitoring. Background research shows that integrating systems for ordering, kitchen updates, and inventory can significantly improve workflow coordination and operational accuracy. This highlights the need for a unified system that connects all key functions within the restaurant. The main problem this project aims to address is the lack of an integrated system that allows real-time communication between the cashier, kitchen staff, and management while also monitoring inventory effectively. Without such a system, restaurants experience operational delays, inconsistencies, and difficulty in tracking performance and resources.

This project will cover the design and development of a web-based system that includes modules for user management, order processing, kitchen workflow updates, and inventory monitoring. It will also discuss the system's objectives, scope, and limitations, methodology, and implementation timeline to ensure a structured and complete approach to solving the identified problem.



## STATEMENT OF THE PROBLEM

Many restaurants currently rely on manual or disconnected systems for handling orders, managing kitchen workflow, and monitoring inventory. This lack of integration leads to inefficiencies such as delayed service, miscommunication between staff, and inaccurate tracking of stock levels.

Specifically, restaurants experience challenges such as slow order processing, difficulty in updating order statuses in real time, and a lack of coordination between cashier and kitchen staff. In addition, poor inventory monitoring can result in stock shortages or overstocking, which affects overall operations and profitability.

This study aims to address the following problems:

1. How can a web-based system improve the efficiency of managing customer orders in a restaurant?
2. How can kitchen staff effectively receive and update order statuses (pending, accepted, preparing, now serving, and completed) in real time?
3. How can the system ensure accurate monitoring and management of inventory levels?
4. How can the system improve communication and coordination among cashier, kitchen staff, admin, and manager?
5. How can the system provide organized records and reports for better decision-making?

## OBJECTIVES OF THE STUDY

### General Objective

To design and develop GrabEat, a web-based restaurant management system that improves order processing, kitchen workflow, and inventory monitoring to enhance operational efficiency.

### Specific Objectives

1. To develop a secure login system with role-based access for admin, cashier, kitchen staff, and manager.
2. To create an order management module that allows efficient input, tracking, and processing of customer orders.
3. To design a kitchen workflow system that enables real-time updating of order statuses (pending, accepted, preparing, now serving, and completed).
4. To implement an inventory monitoring module that tracks stock levels and provides low-stock alerts.



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5. To generate organized reports for sales, orders, and inventory to support decision-making.
6. To improve coordination and communication among restaurant staff through a centralized web-based system.

## SCOPE AND LIMITATION

### Scope

This study focuses on the development of GrabEat, a web-based restaurant management system intended for small to medium-sized restaurants. The system covers key features such as user authentication with role-based access (admin, cashier, kitchen staff, and manager), order management, kitchen workflow updates, and inventory monitoring. It includes order status tracking (pending, accepted, preparing, now serving, and completed), basic reporting, and a centralized dashboard. The system will be accessible through a web browser within the restaurant's operational environment during the project timeline of March to May 2026.

### Limitation

The system is limited to internal restaurant operations and does not include online customer ordering or mobile application integration. It will not support multi-branch management, advanced accounting systems, or external payment gateways. Inventory monitoring will be limited to basic stock tracking and low-stock alerts. Notifications will be system-based only and may not include SMS or email features. The project is developed for academic purposes and may not fully meet enterprise-level standards for security, scalability, and accessibility.

## METHODOLOGY

### Research Design

The study uses a mixed-methods research design, combining qualitative and quantitative approaches. The qualitative approach is used to understand current restaurant processes through observation and interviews, while the quantitative approach is used to evaluate system performance and user responses during testing.



## Data Collection Methods

The data for this study will be collected through:

- Observation of current restaurant ordering and kitchen processes
- Interviews with potential users, such as cashiers and kitchen staff
- Review of related systems and workflows

## Software/Tools Used

The following tools will be used in developing the system:

- Frontend: HTML, CSS, Bootstrap, JavaScript
- Backend: PHP
- Database: MySQL
- Development Environment: XAMPP
- Design Tool: Figma

## Development Process

The study follows the Agile Model, which uses iterative development cycles to continuously improve the system based on feedback.



**Figure 1.** Agile Development Process Model



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Figure 1 shows the Agile Development Process used in developing the GrabEat system. It includes six phases: Requirements, Design, Development, Testing, Deployment, and Review. These phases are repeated in cycles to continuously improve the system based on feedback and performance.

## Requirements

This phase involves gathering and analyzing the system requirements, including features such as order processing, kitchen workflow, and inventory monitoring. Information is collected from observations and interviews to clearly define what the system needs to achieve.

## Design

In this phase, the system structure is planned by creating the database design, system architecture, and user interface prototypes. Wireframes and layouts are developed to visualize how users will interact with the system.

## Development

This phase focuses on coding and building the system modules, including user authentication, order management, kitchen workflow, and inventory monitoring. Each component is developed and integrated step by step.

## Testing

The system is tested to identify errors and ensure all features function correctly. Functional testing and debugging are performed to verify that the system meets the required specifications.

## Deployment

In this phase, the system is implemented in a working environment for actual use or demonstration. The completed system is made accessible to users for operation.

## Review

This final phase involves evaluating the system's performance and collecting user feedback. Improvements and adjustments are made based on the results to enhance system functionality and usability.



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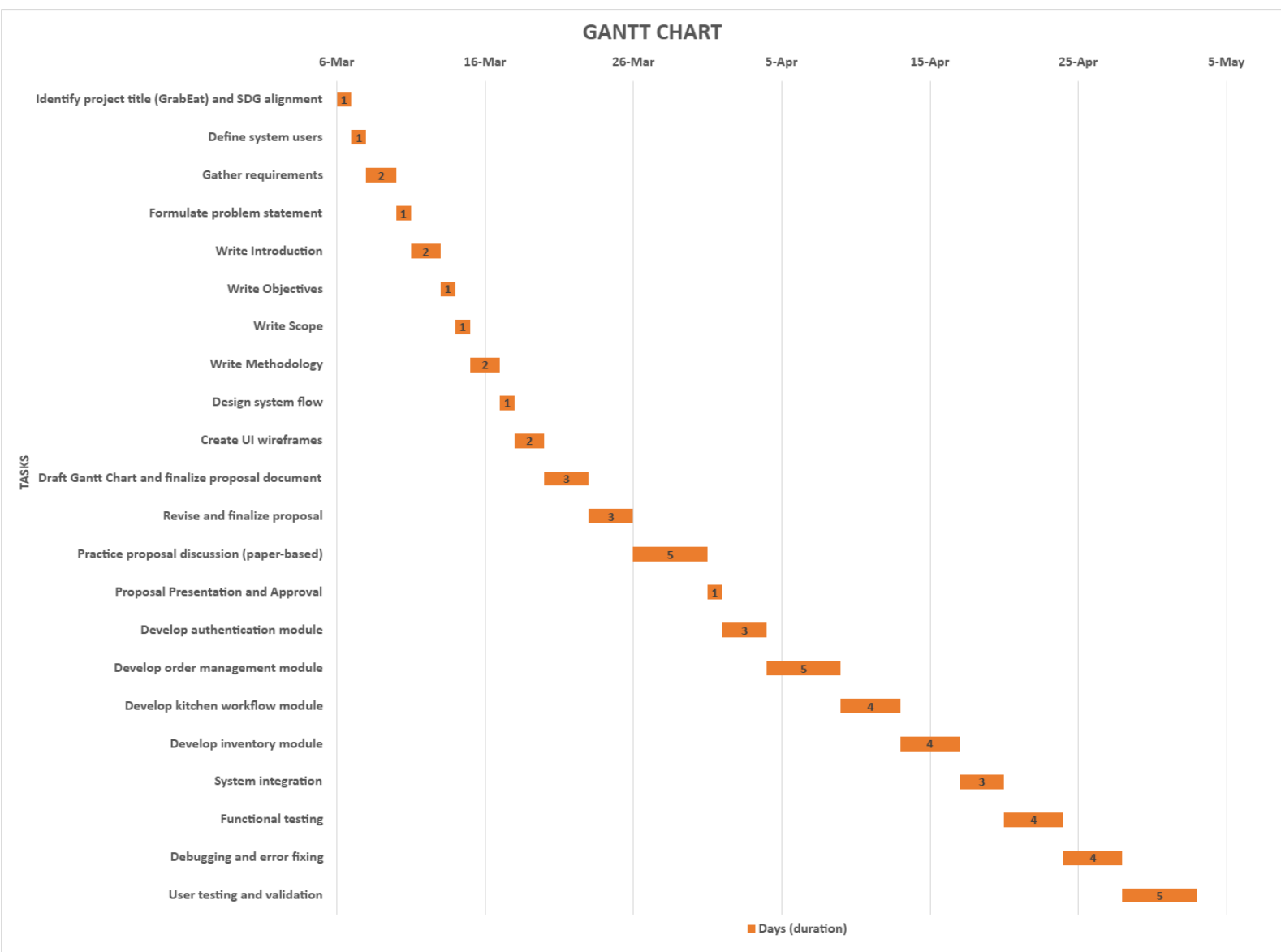
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## Data Analysis and Validation

The system will be analyzed and validated through continuous testing in each Agile cycle, including functional testing, debugging, and user testing. Feedback from users and test results will be used to evaluate the system's effectiveness and ensure it meets the required functionalities.

## GANTT CHART







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The Gantt Chart visually represents the timeline of the GrabEat project, showing the sequence of tasks, their start and end dates, durations, and dependencies. It provides a clear overview of how the project progresses from planning to system development and testing.

| PHASE          | TASKS                                              | START     | END       | DAYS | DEPENDENCY         |
|----------------|----------------------------------------------------|-----------|-----------|------|--------------------|
| Planning       | Identify project title (GrabEat) and SDG alignment | 3/6/2026  | 3/6/2026  | 1    | None               |
| Planning       | Define system users                                | 3/7/2026  | 3/7/2026  | 1    | Title              |
| Analysis       | Gather requirements                                | 3/8/2026  | 3/9/2026  | 2    | Users              |
| Analysis       | Formulate problem statement                        | 3/10/2026 | 3/10/2026 | 1    | Requirements       |
| Analysis       | Write Introduction                                 | 3/11/2026 | 3/12/2026 | 2    | Problem            |
| Analysis       | Write Objectives                                   | 3/13/2026 | 3/13/2026 | 1    | Introduction       |
| Analysis       | Write Scope                                        | 3/14/2026 | 3/14/2026 | 1    | Objectives         |
| Design         | Write Methodology                                  | 3/15/2026 | 3/16/2026 | 2    | Scope              |
| Design         | Design system flow                                 | 3/17/2026 | 3/17/2026 | 1    | Methodology        |
| Design         | Create UI wireframes                               | 3/18/2026 | 3/19/2026 | 2    | Flow               |
| Implementation | Draft Gantt Chart and finalize proposal document   | 3/20/2026 | 3/22/2026 | 3    | All Sections       |
| Implementation | Revise and finalize the proposal                   | 3/23/2026 | 3/25/2026 | 3    | Draft              |
| Testing        | Practice proposal discussion (paper-based)         | 3/26/2026 | 3/30/2026 | 5    | Final Proposal     |
| Deployment     | Proposal Presentation and Approval                 | 3/31/2026 | 3/31/2026 | 1    | All Proposal Tasks |
| Implementation | Develop authentication module                      | 4/1/2026  | 4/3/2026  | 3    | Approval           |
| Implementation | Develop an order management module                 | 4/4/2026  | 4/8/2026  | 5    | Auth               |



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|                |                                 |           |           |   |             |
|----------------|---------------------------------|-----------|-----------|---|-------------|
| Implementation | Develop kitchen workflow module | 4/9/2026  | 4/12/2026 | 4 | Orders      |
| Implementation | Develop an inventory module     | 4/13/2026 | 4/16/2026 | 4 | Kitchen     |
| Implementation | System integration              | 4/17/2026 | 4/19/2026 | 3 | Modules     |
| Testing        | Functional testing              | 4/20/2026 | 4/23/2026 | 4 | Integration |
| Testing        | Debugging and error fixing      | 4/24/2026 | 4/27/2026 | 4 | Testing     |
| Testing        | User testing and validation     | 4/28/2026 | 5/2/2026  | 5 | Debugging   |



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